Ningbo TengLi Testing Co., Ltd

2nd floor, Block B, Ningbo Testing and Certification Base, No. 66
Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang
Tel: 86574-8783 6802
Fax: 86574-8783 5902

Certificate#4703.02

In Situ Temperature Measurement Test Report

For

Torshare Ltd.

(Brand Name: TORSHARE)

5th Minying Road No.3 Industrial Area Shuitian Shiyan Town Bao'an District, Shenzhen, Guangdong, 518108

Model name(s): TAA24-7G-50W-P-TCP[Blank;(MS)]

Type of Luminaire:	2x4 Luminaires for Ambient Lighting of Interior Commercial Spaces
Report Date:	2023-11-21
Prepared By:	Ningbo TengLi Testing Co., Ltd 2nd floor, Block B, Ningbo Testing and Certification Base, No. 66 Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang

Test & Report By:

Engineer: Nick Song

Review By:

Manager: Garman Mo

Note: 1. The results contained in this report pertain only to the tested samples

2. This report does not imply product certification, approval, or endorsement by A2LA, or any agency of the Federal Government.

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1 General

1.1 Product Information:

Model Number	TAA24-7G-50W-P-TCP[Blank;(MS)]	
Remark	The [Blank;(MS)] represents for sensor and function, Blank=without motion sensor, MS= with motion sensor.	
Representative (Tested) Model	TAA24-7G-50W-P-TCP(0%,3500K) TAA24-7G-50W-P-TCP(100%,5000K)	
Model Difference	N/A	
SKU (if available)	N/A	
Type of Luminaire (for integral lamps, list base type and lamp type)	2x4 Luminaires for Ambient Lighting of Interior Commercial Spaces	
LED Manufacturer	Bridgelux Inc.	
LED Model	BXEN-35E-21M-3C BXEN-50E-21M-3C	
LED Model Driver	XZ-SE50B-450103-082063-Y-D	
Dimming	Continuous	
Integral Controls	Yes	
Sample Number	JDE231002-A1	
Date of Receipt	Nov.15,2023	
Luminaire Aperture (for downlights)	--	in.
Luminaire Length	--	mm
Luminaires Width	--	mm
Number of Units (modular products)	N/A	s

1.2 Rated Values:

Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	50W/40W/30W(Power adjustable)
Rated Initial Lamp Lumen	--
Declared CCT	3500K,4000K,5000K(Color Tunable)

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1.3 Standards or methods

The following standards are partly or totally used or referenced for test:

No.	Name
ANSI/UL 1598:2008	**Luminaires**

1.4 Equipment list

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-704	Power Meter	2022-12-29	2023-12-28
ST-R-607	Temperature Tester	2023-01-03	2024-01-02

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2 Test conducted and method

2.1 Ambient Condition

Test was conducted in an ambient temperature of 25±5 ℃. Ambient temperature variations above or below 25 ℃ was subtracted from or added to temperatures recorded at points on the luminaire.
The ambient temperature was measured by a thermocouple which was immersed in 15ml of mineral oil in a glass container.

2.2 Temperature Stabilization

Temperatures were measured after they have stabilized when the test has been running for a minimum of 7.5 hours, or the test has been running for a minimum of 3 hours and three successive reading taken at 15 minutes intervals are with 1 ℃ of another and are not rising.

2.3 Thermocouples

Type K thermocouple was used for temperature measurement. The thermocouple was 0.05mm2(30AWG), and complied with the requirements specified in ASTM MNL 12 and limits of error specified in NIST ITS 90 and ISA MC96.1.

2.4 Thermocouples contact

Thermocouples were in contact with the TMP LED location described in LM-80 test report. In order to gain the maximum temperature, if appropriate, more than one thermocouple were contact in these locations. For details information, please refer to clause 3.3 for the photo of thermocouple contact.

3 Test Results

Test date	2023-11-17	Test Ambient	25.1 ℃
Sample No.		LED Package Model	
JDE231002-A1(0%,3500K)		BXEN-35E-21M-3C	
LED driver of Each Lamp	Output voltage V	Measured LED working current (Max.) mA	
1	43.2	41.11	
Sample No.		LED Package Model	
JDE231002-A1(100%,5000K)		BXEN-50E-21M-3C	
LED driver of Each Lamp	Output voltage V	Measured LED working current (Max.) mA	
1	43.2	41.05	

3.1 Test Data:

0%,3500K

Input Vol.	120.0V	Input Current	0.3970A	Input Wattage	47.40W	Temperature stabilization time:	500 min

No.	Temperature (°C)		No.	Temperature (°C)		No.	Temperature (°C)	
	Measured	Corrected at 25°C		Measured	Corrected at 25°C		Measured	Corrected at 25°C
1	40.1	40.0	2	40.4	40.3	3	40.3	40.2
The highest in-situ measured temperature LED is 40.3°C								

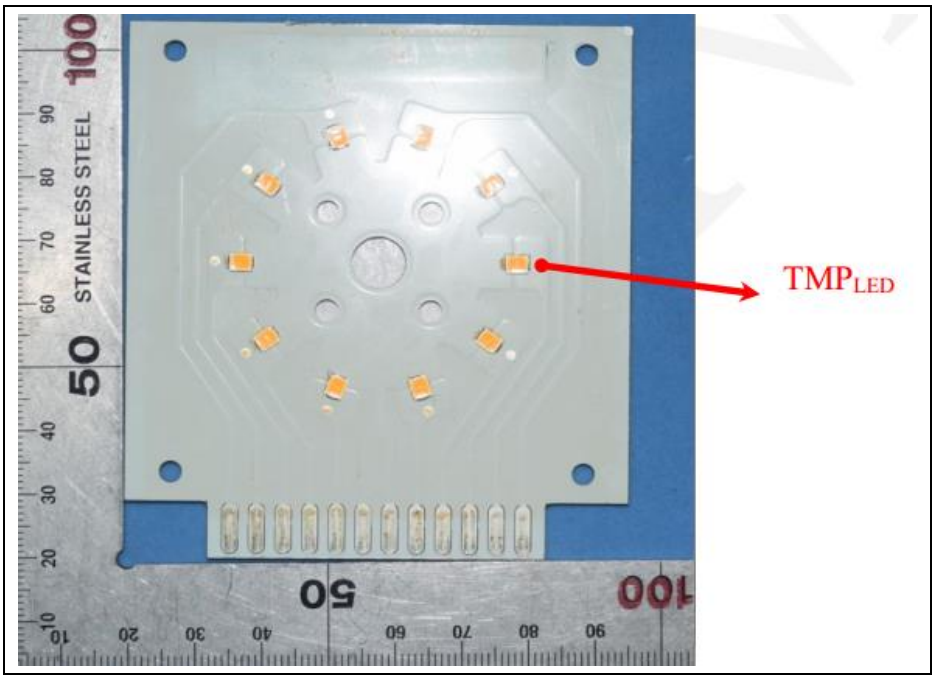
100%,5000K

Input Vol.	120.0V	Input Current	0.3955A	Input Wattage	47.25W	Temperature stabilization time:	500 min

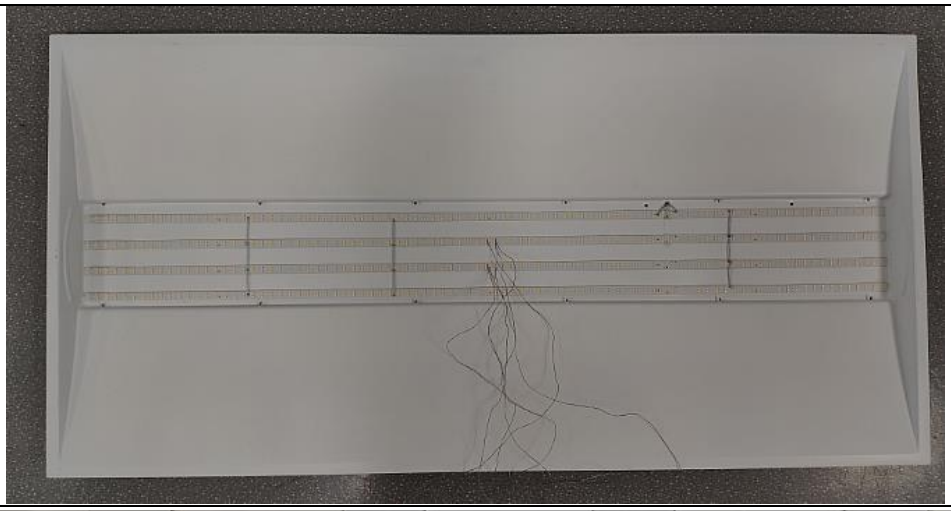
No.	Temperature (°C)		No.	Temperature (°C)		No.	Temperature (°C)	
	Measured	Corrected at 25°C		Measured	Corrected at 25°C		Measured	Corrected at 25°C
1	39.9	39.8	2	39.6	39.5	3	39.8	39.7
The highest in-situ measured temperature LED is 39.8°C								

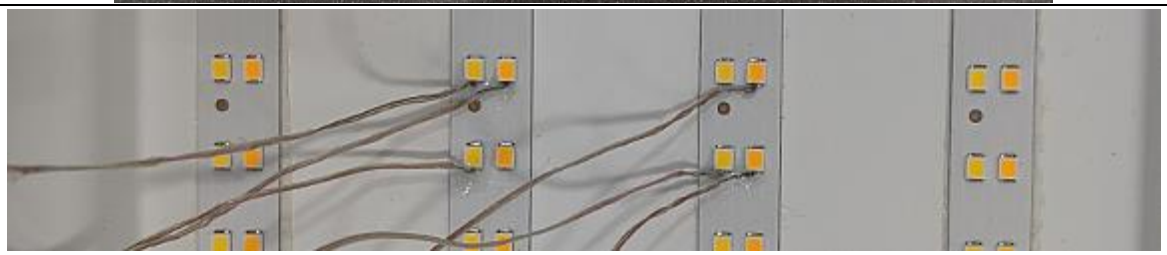
3.2 Test Photo:

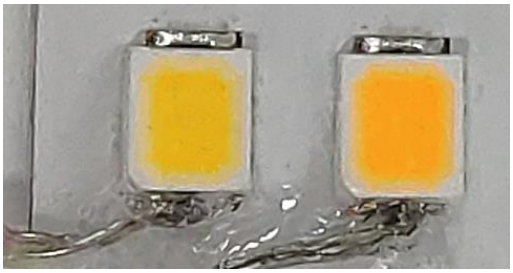
Ts Position:



Thermocouple Location on Temperature Measurement Point (TMP):







Results

Time (t) at which to estimate lumen maintenance (hours):	50,000
Lumen maintenance at time (t) (%):	89.57%
Reported L70 (hours):	>60000

Time (t) at which to estimate lumen maintenance (hours):	36,000
Lumen maintenance at time (t) (%):	92.54%
Reported L90 (hours):	48,000

3.3 Test Data of LED Driver:

0%,3500K

Input Vol.	120.0V	Input Current	0.3970A	Input Wattage	47.40W	Temperature stabilization time:	500 min
No	Measured TC Temperature (°C)			Temperature Limited of Life ≥ 50000 hours			
	Measured		Corrected at 25°C				
1	42.6		42.5	65			

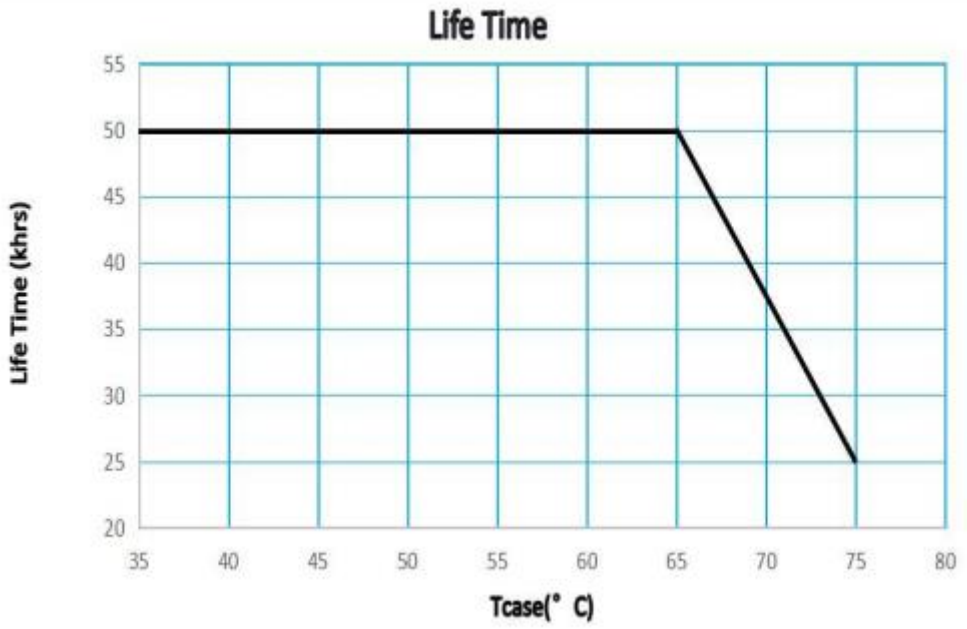
100%,5000K

Input Vol.	120.0V	Input Current	0.3955A	Input Wattage	47.25W	Temperature stabilization time:	500 min
No	Measured TC Temperature (°C)			Temperature Limited of Life ≥ 50000 hours			
	Measured		Corrected at 25°C				
1	42.4		42.3	65			

3.4 Test Photo:

Thermocouple Location on Temperature Measurement Point (TMP):

Driver Lifetime: 驱动器寿命:	at Tc ≤ 65°C （Tc at CY3）,50000 hours; 在Tc点≤ 65°C（Tc at CY3），50000 小时;


Life Time
Life Time (khrs)
55
50
45
40
35
30
25
20
35
40
45
50
55
60
65
70
75
80
Tcase(° C)

4. Product Photo

****** END OF THE TEST REPORT******